2015-2016 – Econ590
Topics in Macroeconomics

Lecture 2 B : Structural VARs
An AD-AS Example

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0. Introduction

- Here we illustrate the logic of shocks identification is structural VARs.
- As the model used for identification, we choose the AD/AS model of the Neoclassical synthesis, that was the standard workhorse of macroecomics until the mid 70's, and that is still the common wisdom among many politicians and journalists.
- IS-LM and AD-AS models
- General Equilibrium without explicit micro-foundations of individual decisions and description of markets organization

1. A First Look at the AD-AS Model

- AD : Level of aggregate demand as a function of the general price level (M = money supply, G=govt expenditures)

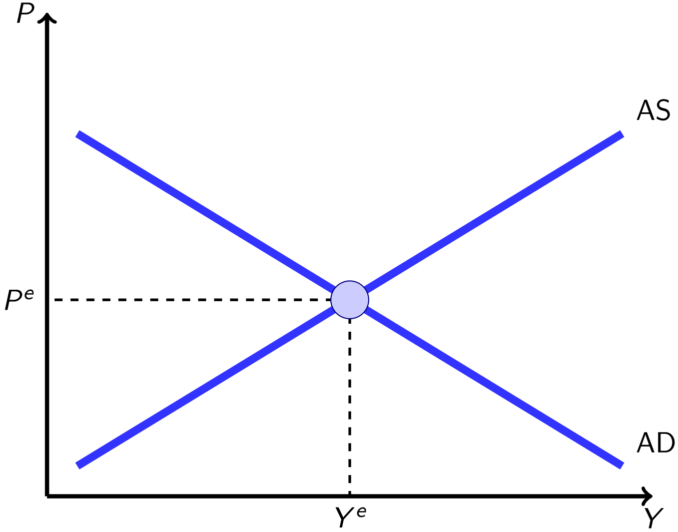
$$P = a_0 - a_1 Y + a_2 M + a_3 G \tag{1}$$

- AS : Level of aggregate supply as a function of the general price level (Q = productivity)

$$Y = b_0 + b_1 P + b_2 Q \tag{2}$$

- Note : Y = value added = income = expenditures (regarding only final goods)

FIGURE 1 – Equilibrium of the AD-AS Model



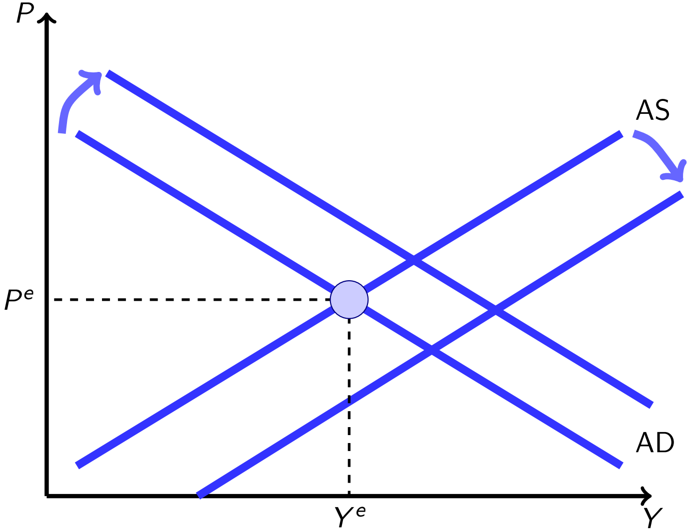
1. A First Look at the AD-AS Model

- One can compute the solution and multipliers

$$
\begin{aligned}
P &= \left(\frac{a_0 - a_1 b_0}{1 + a_1 b_1}\right) + \left(\frac{a_2}{1 + a_1 b_1}\right) M + \left(\frac{a_3}{1 + a_1 b_1}\right) G - \left(\frac{a_1 b_2}{1 + a_1 b_1}\right) Q \\
Y &= \left(\frac{b_0 + b_1 a_0}{1 + a_1 b_1}\right) + \left(\frac{b_1 a_2}{1 + a_1 b_1}\right) M + \left(\frac{b_1 a_3}{1 + a_1 b_1}\right) G + \left(\frac{b_2}{1 + a_1 b_1}\right) Q
\end{aligned}
$$

- The model is mainly used for policy evaluation : $\frac{\partial Y}{\partial G}$, $\frac{\partial Y}{\partial M}$, $\frac{\partial Y}{\partial Q}$, $\frac{\partial P}{\partial M}$, *etc...* (oil price shock, monetary expansion, ...)
- Most of the debate was then the size and signs of the multipliers.
- This model have foundations, although not micro-foundations : IS-LM for AD and the functioning of the labor market for AS.

FIGURE 2 – Shocks and Policies in the AD-AS Model



2. Identification and Economic Interpretation

- Let's use a Blanchard & Quah (AER 1989) (BQ) type of analysis to evaluate the relative importance of "supply" and "demand" shock
- The idea is to decompose any movement of the economy as the consequence of 2 orthogonal shocks : a demand shock and a supply one.

2. Identification and Economic Interpretation

- Assume that the model economy is the following AD-AS :

$$\begin{cases} P = -\alpha Y + \varepsilon^D & (AD) \\ P = \beta Y - \varepsilon^S & (AS) \end{cases}$$

- α and β are positive constants
- Shocks are zero-mean stochastic variables

2. Identification and Economic Interpretation

- Remark : How do we obtain such a linear model in which the non-stochastic solution is $Y = P = 0$?

$$\begin{cases} \mathcal{P} &= A \ \mathcal{Y}^{-\alpha} \ e^{\varepsilon^D} & (AD) \\ \mathcal{P} &= B \ \mathcal{Y}^{\beta} \ e^{-\varepsilon^S} & (AS) \end{cases}$$

- Compute non-stochastic solution :

$$\begin{cases} \overline{\mathcal{Y}} &= \left(\frac{A}{B}\right)^{\frac{1}{\alpha+\beta}} \\ \overline{\mathcal{P}} &= A^{\frac{\beta}{\alpha+\beta}} B^{\frac{\alpha}{\alpha+\beta}} \end{cases}$$

2. Identification and Economic Interpretation

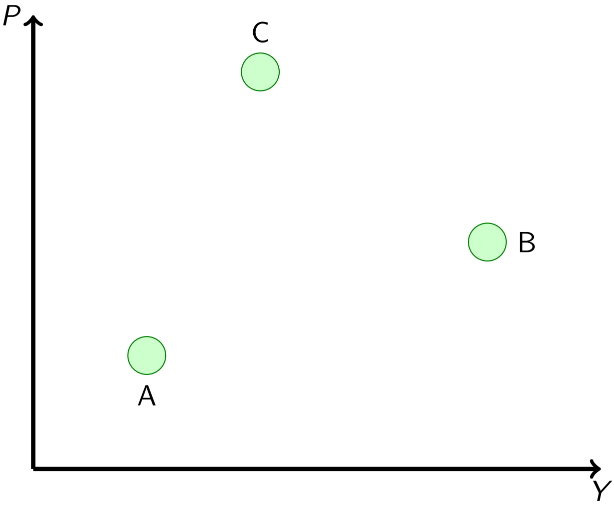
- Denote $X = \log \mathcal{X} - \log \overline{\mathcal{X}}$
- We then obtain

$$\begin{cases} P = -\alpha Y + \varepsilon^D & (AD) \\ P = \beta Y - \varepsilon^S & (AS) \end{cases}$$

- Assume that we know α and β.
- Then, one can identify demand and supply shocks (namely ε^D and ε^S)

2. Identification and Economic Interpretation

FIGURE 3 – Observation : The economy went from A to B and C



2. Identification and Economic Interpretation

Figure 4 – We aim at putting names (stories) on those green arrows

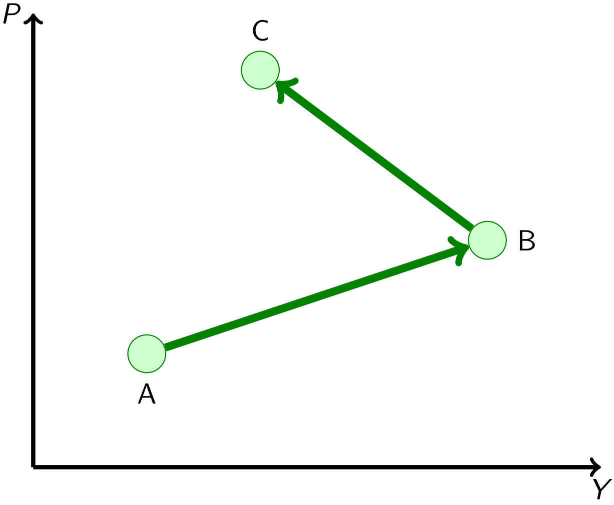
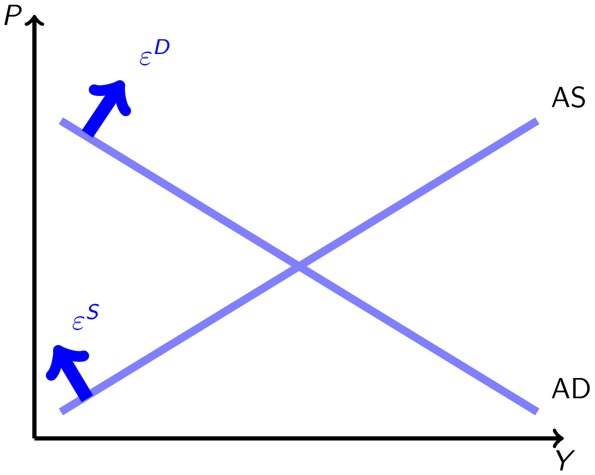
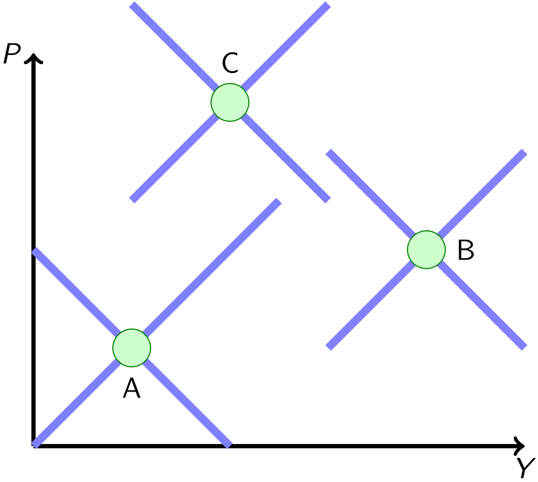


FIGURE 5 – The AD-AS model provides us with a theory of economic fluctuations (the green arrows) with the help of the blue shifts



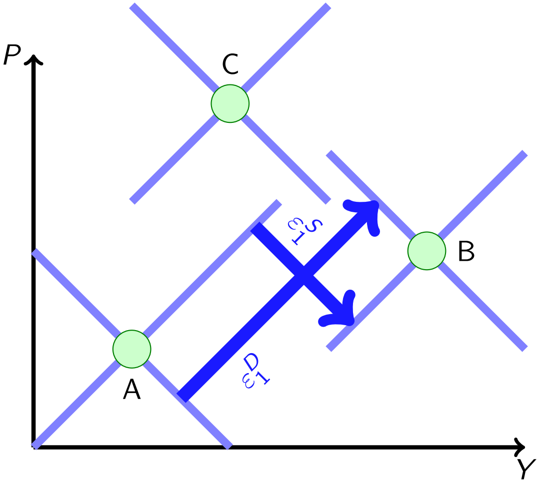
2. Identification and Economic Interpretation

FIGURE 6 – Each Observation is at the crossing of one AD and one AS curve



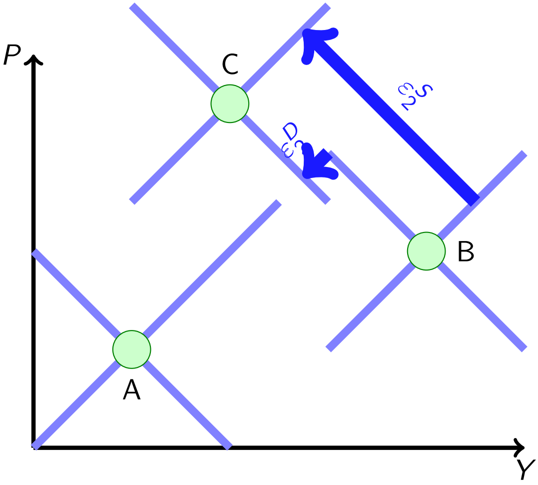
2. Identification and Economic Interpretation

FIGURE 7 – This is the structural interpretation of the move from A to B



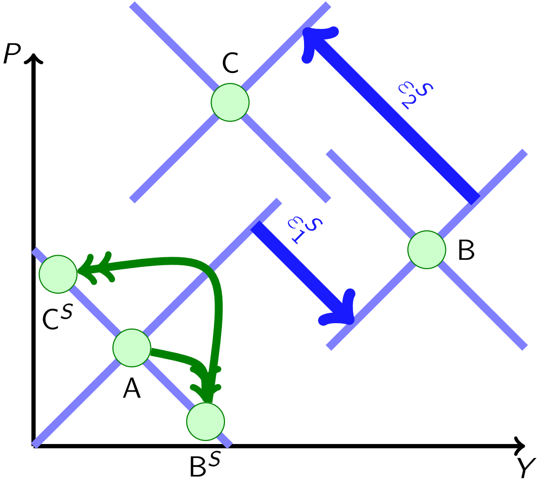
2. Identification and Economic Interpretation

FIGURE 8 – This is the structural interpretation of the move from B to C



2. Identification and Economic Interpretation

FIGURE 9 – Counterfactual : What would have happen absent of demand shocks



2. Identification and Economic Interpretation

- The algebra is even more simple. If one solves the model

$$\begin{cases} P = -\alpha Y + \varepsilon^D & (AD) \\ P = \beta Y - \varepsilon^S & (AS) \end{cases}$$

one gets

$$\begin{cases} P = \frac{\beta}{\alpha+\beta}\varepsilon^D - \frac{\alpha}{\alpha+\beta}\varepsilon^S \\ Y = \frac{1}{\alpha+\beta}\varepsilon^D + \frac{1}{\alpha+\beta}\varepsilon^S \end{cases}$$

- When one observes Y and P, this is a set of 2 equations with 2 unknowns, ε^D and ε^S $\rightsquigarrow$ one can recover the structural shocks.
- The problem is that in the real world, we do not know α and β

2. Identification and Economic Interpretation

- One way could be to estimate each of the two equations using instrumental variables (oil price when estimating AD, money supply or Gvt expenditures when estimating AS)
- But it is very unlikely that this very simple and static model captures a significant part of the economy variance

3. A Dynamic Model

- Assume that the economy is best described by the following dynamic model :

$$\begin{cases} P_t &= \alpha_0^D Y_t + \alpha_1^D Y_{t-1} + \alpha_2^D Y_{t-2} + \cdots + \alpha_N^D Y_{t-N} \\ & + \beta_1^D P_{t-1} + \beta_2^D P_{t-2} + \cdots + \beta_N^D P_{t-N} + \varepsilon_t^D \qquad (AD) \\ P_t &= \alpha_0^S Y_t + \alpha_1^S Y_{t-1} + \alpha_2^S Y_{t-2} + \cdots + \alpha_N^S Y_{t-N} \\ & + \beta_1^S P_{t-1} + \beta_2^S P_{t-2} + \cdots + \beta_N^S P_{t-N} + \varepsilon_t^S \qquad (AS) \end{cases}$$

- Demand and Supply shocks are independent.

3. A Dynamic Model

- Consider the simple dynamic model

$$X_t = \rho X_{t-1} + \varepsilon_t \qquad (1)$$

with $0 < \rho < 1$ and ε *iid*, with $E(\varepsilon_t) = 0$, $V(\varepsilon_t) = \sigma^2$
- (1) is the autoregressive (AR) representation of the model.
- One can write a moving average (MA) representation, which expresses X_t as a function of the current and past values of ε.

$$\begin{aligned} X_t &= \rho X_{t-1} + \varepsilon_t \\ &= \rho(\rho X_{t-2} + \varepsilon_{t-1}) + \varepsilon_t \\ &= \rho^2 X_{t-2} + \rho\varepsilon_{t-1} + \varepsilon_t \\ &= \rho^3 X_{t-3} + \rho^2\varepsilon_{t-2} + \rho\varepsilon_{t-1} + \varepsilon_t \\ &= \rho^n X_{t-n} + \rho^{n-1}\varepsilon_{t-n+1} + \cdots + \rho\varepsilon_{t-1} + \varepsilon_t \end{aligned}$$

- When $n \to \infty$, as $0 < \rho < 1$

$$X_t = \sum_{i=0}^{\infty} \rho^i \varepsilon_{t-i}$$

3. A Dynamic Model

- Making use of the lag operator notation : $LX_t = X_{t-1}$, $L^i X_t = X_{t-i}$, $i \ \in \mathcal{Z}$

$$
\begin{aligned}
X_t &= \rho X_{t-1} + \varepsilon_t \\
X_t &= \rho L X_t + \varepsilon_t \\
(1-\rho L) X_t &= \varepsilon_t \\
X_t &= \frac{1}{1-\rho L} \varepsilon_t \\
X_t &= (1 + \rho L + \rho^2 L^2 + \cdots + \rho^n L^n + \cdots) \varepsilon_t \\
X_t &= \varepsilon_t + \rho \varepsilon_{t-1} + \cdots + \rho^{n-1} \varepsilon_{t-n+1} + \rho^n \varepsilon_{t-n} + \cdots
\end{aligned}
$$

- More generally, a univariate MA(∞) is denoted :

$$
X_t = a_0 \varepsilon_t + a_1 \varepsilon_{t-1} + \cdots + a_{n-1} \varepsilon_{t-n+1} + a_n \varepsilon_{t-n} + \cdots
$$

3. A Dynamic Model

- One can use the MA representation of the model to compute various indicators.

  Impulse response function : what is the dynamic effect of a shock ? Assume that for $t = -\infty$ to 0, $\varepsilon_t = 0$, such that $X_0 = 0$.
- The shock is $\varepsilon_1 = 1$, with $\varepsilon_t = 0$ for $t > 1$.
- Then $X_1 = 1$, $X_2 = \rho$, $X_3 = \rho^2$, etc...
- Note that the IRF is given by the coefficients of the MA representation.

Forecast Error Variance : The mathematical expectation of X_{t+1} based on the information of period t is denoted $E_t(X_{t+1})$.

$$\begin{aligned} E_t(X_{t+1}) &= E_t(\rho X_t + \varepsilon_{t+1}) \\ &= \rho \underbrace{E_t X_t}_{X_t} + \underbrace{E_t \varepsilon_{t+1}}_{0} \end{aligned}$$

- The one-step forecast error if X_t is $FE_1 = X_{t+1} - E_t(X_{t+1}) = \varepsilon_{t+1}$.
- $E_t(FE_1) = 0$ and $V(FE_1) = \sigma^2$
- Similarly,
  $FE_k = X_{t+k} - E_t(X_{t+k}) = \rho^{k-1}\varepsilon_{t+1} + \rho^{k-2}\varepsilon_{t+2} + \cdots + \varepsilon_{t+k}$,
  $E_t(FE_k) = 0$ and
  $V(FE_k) = (\rho^{2(k-1)} + \rho^{2(k-2)} + \cdots + \rho^2 + 1)\sigma^2$
- Note again that these statistics are functions of the MA coefficients.
- For example, $V(FE_k) = (a_k^2 + a_{k-1}^2 + \cdots + a_1^2 + a_0^2)\sigma^2$

4. VAR and VMA Representations of the Model

- Assume that the economy is best described by the following dynamic model :

$$\begin{cases} P_t & = & \alpha_0^D Y_t + \alpha_1^D Y_{t-1} + \alpha_2^D Y_{t-2} + \cdots + \alpha_N^D Y_{t-N} \\ & + & \beta_1^D P_{t-1} + \beta_2^D P_{t-2} + \cdots + \beta_N^D P_{t-N} + \varepsilon_t^D & (AD) \\ P_t & = & \alpha_0^S Y_t + \alpha_1^S Y_{t-1} + \alpha_2^S Y_{t-2} + \cdots + \alpha_N^S Y_{t-N} \\ & + & \beta_1^S P_{t-1} + \beta_2^S P_{t-2} + \cdots + \beta_S^P P_{t-N} + \varepsilon_t^S & (AS) \end{cases}$$

- making use of the lag operator notation :
$LX_t = X_{t-1}$, $L^i X_t = X_{t-i}$, $i \ \in \mathcal{Z}$

4. VAR and VMA Representations of the Model

- and rearranging terms

$$\begin{cases} Y_t &= (\alpha_1^Y + \alpha_2^Y L + \cdots + \alpha_N^Y L^{N-1}) Y_{t-1} \\ &+ (\beta_1^Y + \beta_2^Y L + \cdots + \beta_N^Y L^{N-1}) P_{t-1} + \gamma_D^Y \varepsilon_t^D + \gamma_S^Y \varepsilon_t^S \\ P_t &= (\alpha_1^P + \alpha_2^P L + \cdots + \alpha_N^P L^{N-1}) Y_{t-1} \\ &+ (\beta_1^P + \beta_2^P L + \cdots + \beta_N^P L^{N-1}) P_{t-1} + \gamma_D^P \varepsilon_t^D + \gamma_S^P \varepsilon_t^S \end{cases}$$

- or equivalently

$$X_t = \widehat{A}(L) X_{t-1} + B\varepsilon_t$$

with $X_t = (Y_t, P_t)'$ and $\varepsilon_t = (\varepsilon_t^D, \varepsilon_t^S)$

- This is the VAR (Vectorial Auto Regressive) representation of the equilibrium.

4. VAR and VMA Representations of the Model

- It is convenient to work with the VMA (Vectorial Moving Average) representation

$$X_t = \frac{B}{I - \widehat{A}(L)L}\varepsilon_t$$

or

$$X(t) = \sum_{j=0}^{\infty} A(j)\varepsilon_{t-j}$$

with $\mathrm{Var}(\varepsilon_t) = I$ and

$$A(j) = \begin{pmatrix} a_{11}(j) & a_{12}(j) \\ a_{21}(j) & a_{22}(j) \end{pmatrix}$$

- This dynamic model, if derived from a structural one, puts a lot of restrictions on the sequence of $A(\cdot)$

5. Impulse Response Function (IRF), Variance decomposition and Historical decomposition

- Here I derive some summary statistics from the VMA representation
- Let us consider output. We have

$$Y_t = \sum_{j=0}^{\infty} a_{11}(j)\varepsilon_{t-j}^{D} + \sum_{j=0}^{\infty} a_{12}(j)\varepsilon_{t-j}^{S}$$

- The IRF to a demand shock is $\{a_{11}(0), a_{11}(1), a_{11}(2), ...\}$ and the IRF to a supply shock is $\{a_{12}(0), a_{12}(1), a_{12}(2), ...\}$

5. Impulse Response Function (IRF), Variance decomposition and Historical decomposition

- The Forecast Error in predicting Y at horizon 1 is

$$Y_{t+1} - E_t Y_{t+1} = a_{11}(0)\varepsilon^D_{t+1} + a_{12}(0)\varepsilon^S_{t+1}$$

and the share of the variance of FE at horizon 1 attributable to the demand shock is

$$\frac{a^2_{11}(0)}{a^2_{11}(0) + a^2_{12}(0)}$$

(the variances of the structural shocks is normalized to 1).

- At horizon k, this share is

$$\frac{\sum_{j=0}^{k} a^2_{11}(j)}{\sum_{j=0}^{k} a^2_{11}(k) + \sum_{j=0}^{k} a^2_{12}(j)}$$

5. Impulse Response Function (IRF), Variance decomposition and Historical decomposition

- Historical decomposition : what would have happen if only demand or supply shocks have been there ?

$$Y_t^D = \sum_{j=0}^{\infty} a_{11}(j)\varepsilon_{t-j}^D$$

$$Y_t^S = \sum_{j=0}^{\infty} a_{12}(j)\varepsilon_{t-j}^S$$

6. The Need For Identification Assumptions

- Let us estimate a VAR model with Y and P.

$$X_t = \widetilde{A}(L)X_{t-1} + \nu_t$$

with $\text{Var}(\nu) = \Omega$ and $C(0) = I$ by normalization.

- Note that the νs are different from the ϵs (they are an unknown linear combination of the ϵs)
- From this estimated VAR form, one can recover the following *non structural* (or *reduced form)* VMA representation

$$X(t) = \sum_{j=0}^{\infty} C(j)\nu_{t-j}$$

- How can ν be cut into two orthogonal pieces that we will label demand and supply shocks ?

6. The Need For Identification Assumptions

- Compare this VMA representation with the *structural* one

$$X(t) = \sum_{j=0}^{\infty} A(j)\varepsilon_{t-j}$$

- As the two equations are representations of the same model,

$$\nu = A(0)\varepsilon \text{ and } A(j) = C(j)A(0) \text{ for } j > 0.$$

- Estimation gives us C.
- Once we know $A(0)$, we have everything. We have therefore 4 unknowns : $a_{11}(0), a_{12}(0)$, $a_{21}(0)$ and $a_{22}(0)$.

6. The Need For Identification Assumptions

- How do we get $A(0)$? First, if $\nu = A(0)\varepsilon$, then ν and $A(0)\varepsilon$ have the same variance-covariance matrix.
- The one of ν is the Ω (estimated). The one of ε is I by assumption.
- Therefore, one has

$$V(A(0)\varepsilon) = V(\nu) \Longleftrightarrow A(0)A(0)' = \Omega$$

or

$$\begin{pmatrix} a_{11}(0) & a_{12}(0) \\ a_{21}(0) & a_{22}(0) \end{pmatrix} \times \begin{pmatrix} a_{11}(0) & a_{12}(0) \\ a_{21}(0) & a_{22}(0) \end{pmatrix}' = \begin{pmatrix} \omega_{11}(0) & \omega_{12}(0) \\ \omega_{12}(0) & \omega_{22}(0) \end{pmatrix}$$

- This gives us 3 equations (because Ω and $A(0)A(0)'$ are symmetrical) for 4 unknowns (the 4 coefficients of $A(0)$)

6. The Need For Identification Assumptions

- We need one identifying assumption, that will allow us to separate aggregate demand shocks from aggregate supply ones.
- This last condition cannot come from the math. It has to be a restriction imposed by the economist, based on some "reasonable" property of the economy.

- Here only one extra restriction is needed because we have a 2-variables VAR. It could be more in larger models.
- This restriction should come from a model.
- Blanchard-Quah proposed the following restriction : *Only supply shocks affect output in the long run* or in other words *Demand shocks do not affect output in the long run*.
- The long run effect of a demand shock is $a_{11}(\infty)$
- But $A(\infty) = C(\infty)A(0)$ or

$$\begin{pmatrix} a_{11}(\infty) & a_{12}(\infty) \\ a_{21}(\infty) & a_{22}(\infty) \end{pmatrix} = \begin{pmatrix} c_{11}(\infty) & c_{12}(\infty) \\ c_{21}(\infty) & c_{22}(\infty) \end{pmatrix} \times \begin{pmatrix} a_{11}(0) & a_{12}(0) \\ a_{21}(0) & a_{22}(0) \end{pmatrix}$$

- The fourth restriction is therefore

$$c_{11}(\infty)a_{11}(0) + c_{21}(\infty)a_{21}(0) = 0$$

- Recall that the $c_{ij}(\infty)$ are known (from estimation).
- We can therefore compute $A(0)$.
- Once we have $A(0)$, and the estimated VAR, we can compute IRF to shocks and Forecast Error Variance decomposition

6. The Need For Identification Assumptions

Some Extra Difficulties

- Here, I have assumed that both the model and the estimated VAR could be written and estimated as

$$X_t = \widehat{A}(L)X_{t-1} + B\varepsilon_t$$

- Both theory and estimation could imply that there are some constants or trends in this expression

$$X_t = \widehat{A}(L)X_{t-1} + B\varepsilon_t + K_1 + K_2 t$$

- There is also an issue about the best way to specify the VAR if the series are non stationary (ie if shocks have permanent effect, which is the case here). It might be more efficient to specify the VAR in difference

$$(1-L)X_t = (1-L)\widehat{A}(L)X_{t-1} + B\varepsilon_t + K_1$$

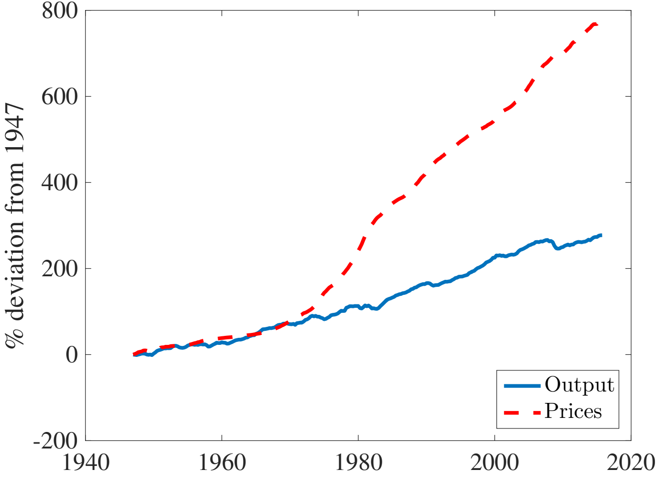
- In each of these case, although the algebra is more tedious, the same results than the one I have presented here apply.

7. Data

- Data : US 1947Q1-2015Q4 quarterly data
- Output is Real GDP per capita, Prices series is the GNP deflator.
- With some abuse of the interpretation of the AD-AS model, we consider not P and Y but ΔP and ΔY.
- Take 12 lags in the VAR

7. Data

Figure 10 – US Output and Prices, 1947Q1-2015Q4



7. Data

FIGURE 11 – US Growth Rates of Output and Prices, 1947Q1-2015Q4

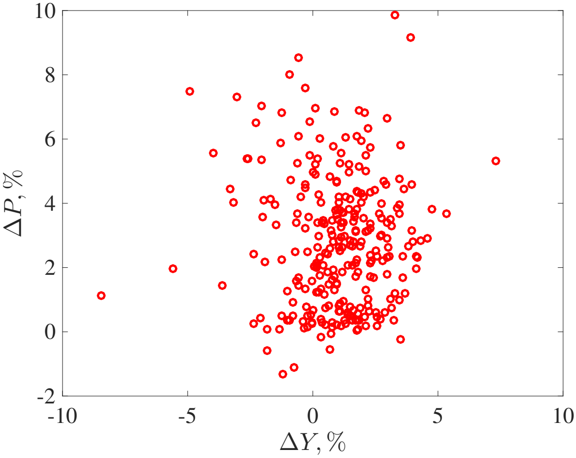
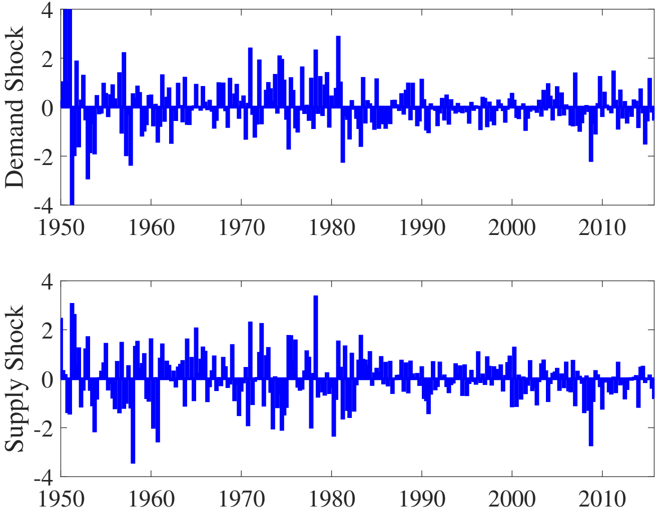
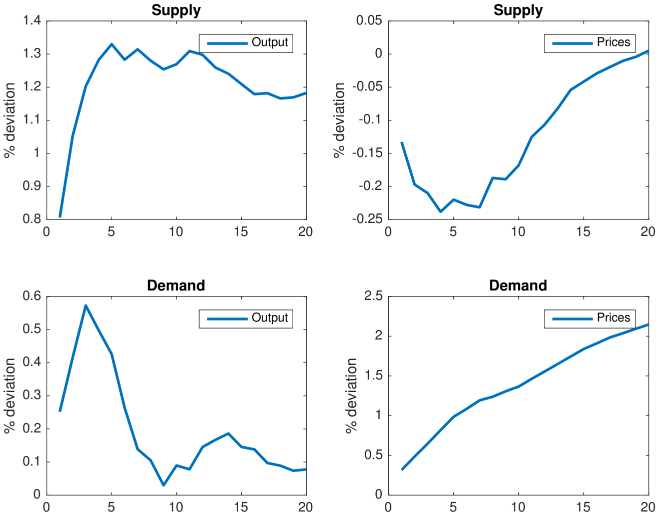


FIGURE 12 – Estimated Shocks



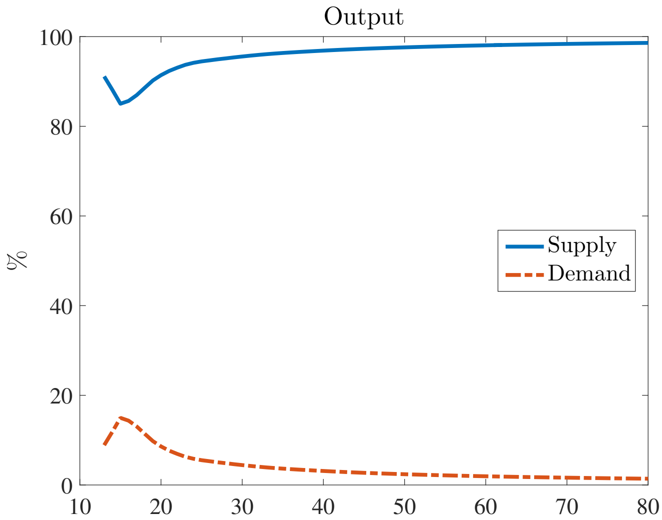
8. Results : IRF and Variance Decomposition

FIGURE 13 – IRF



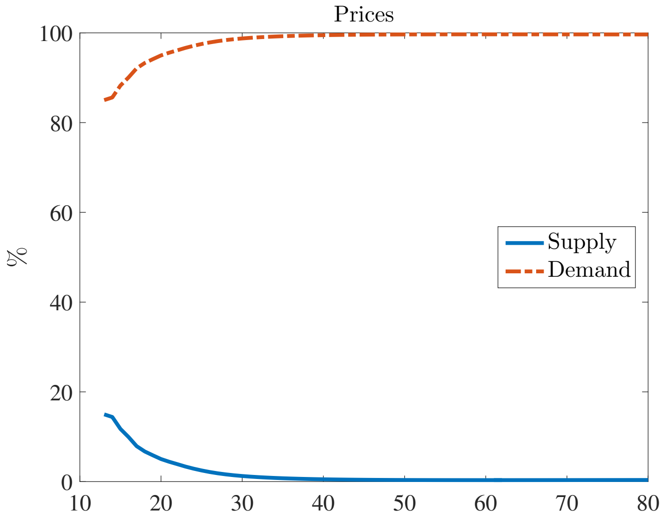
8. Results : IRF and Variance Decomposition

Figure 14 – Output FE Variance Decomposition



8. Results : IRF and Variance Decomposition

FIGURE 15 – Prices FE Variance Decomposition

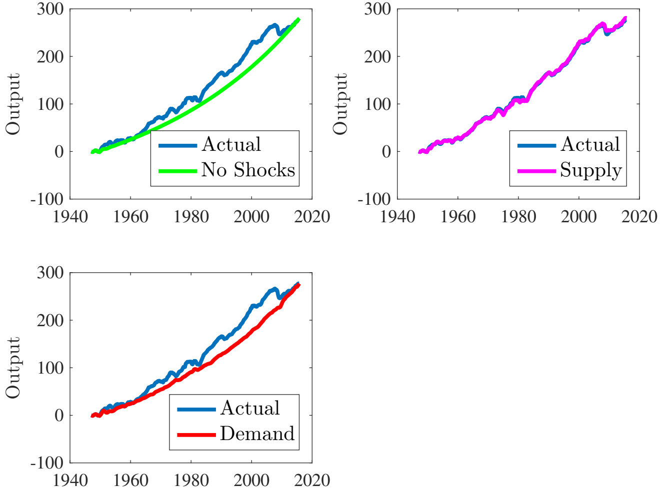


8. Results : Historical Decomposition

- We use the following color code :
  - **Blue : actual series**
  - **Green : the series with no shocks**
  - **Pink : the series with only the supply shocks**
  - **Red : the series with only the demand shocks**

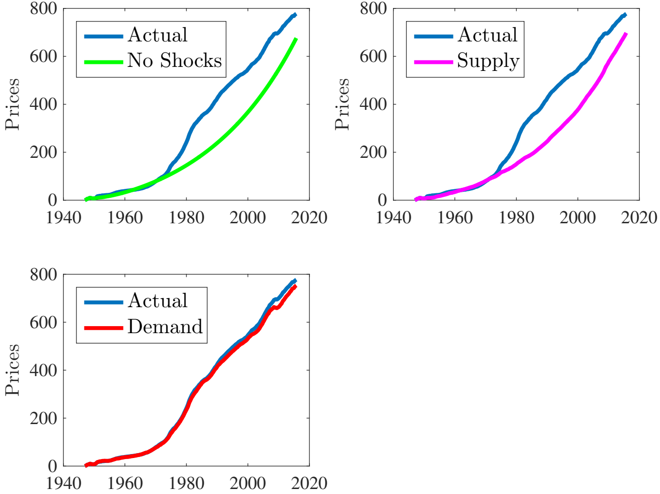
8. Results : Historical Decomposition

FIGURE 16 – Whole Sample



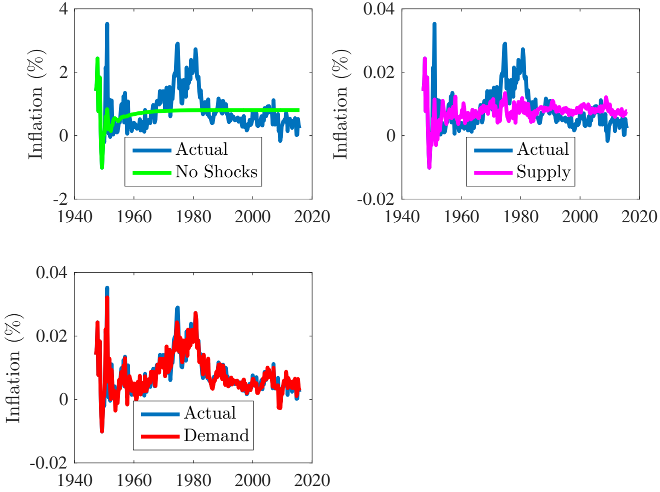
8. Results : Historical Decomposition

FIGURE 17 – Whole Sample



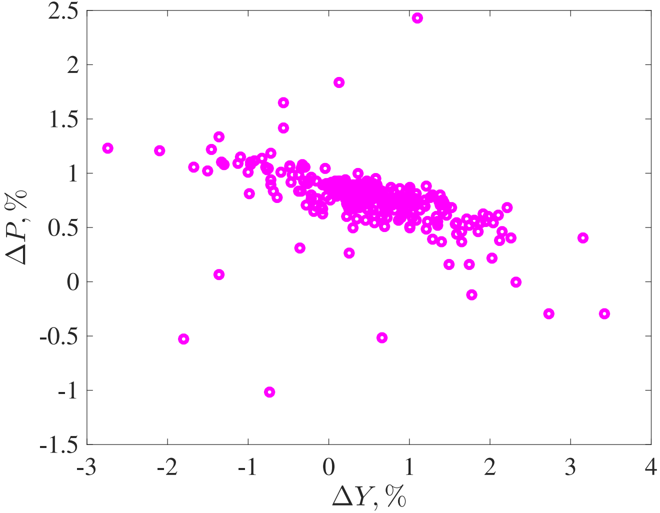
8. Results : Historical Decomposition

FIGURE 18 – Whole Sample



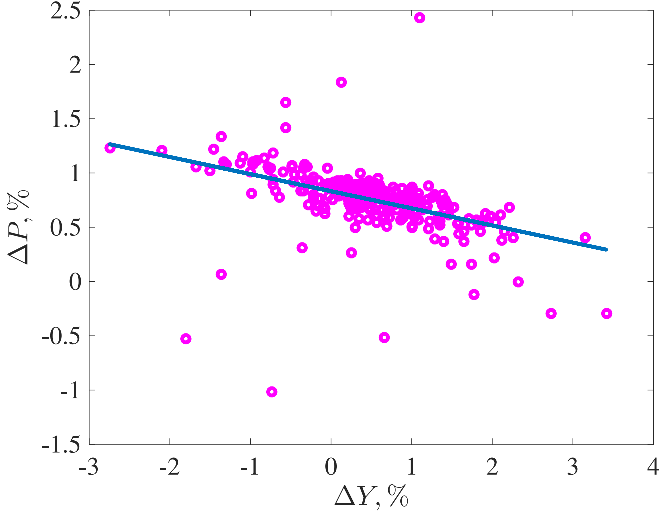
8. Results : Historical Decomposition

FIGURE 19 – Whole Sample - Supply Shocks Only



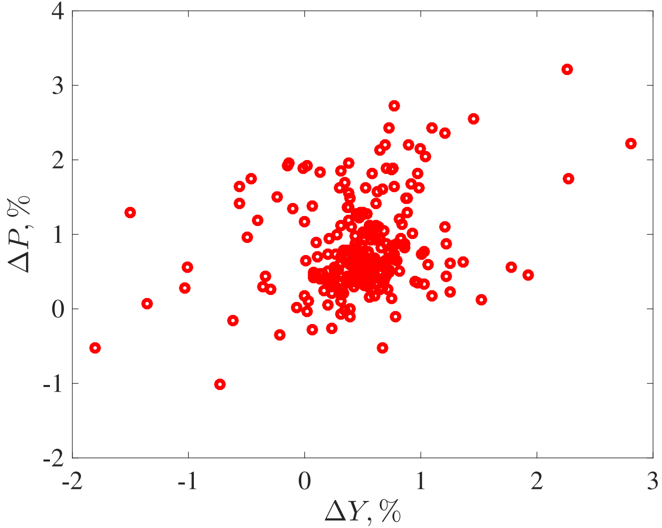
8. Results : Historical Decomposition

FIGURE 20 – Whole Sample - Supply Shocks Only



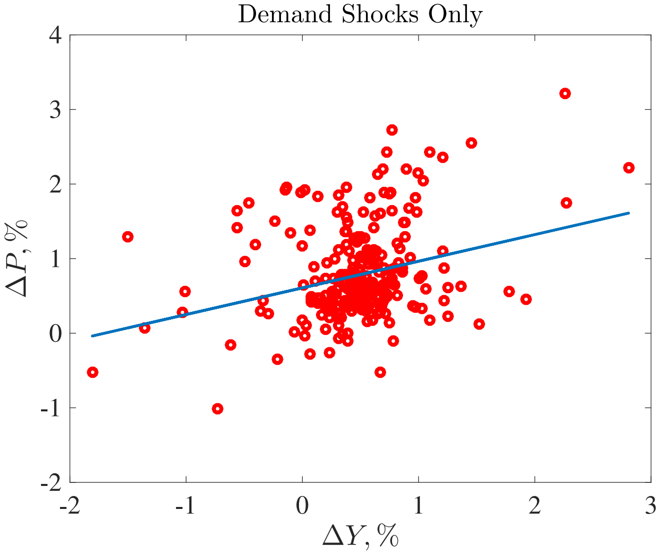
8. Results : Historical Decomposition

FIGURE 21 – Whole Sample - Demand Shocks Only



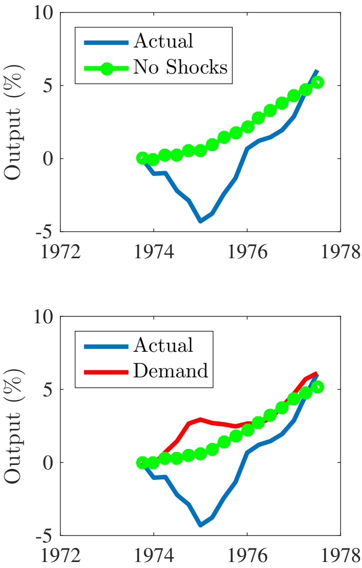
8. Results : Historical Decomposition

FIGURE 22 – Whole Sample - Demand Shocks Only



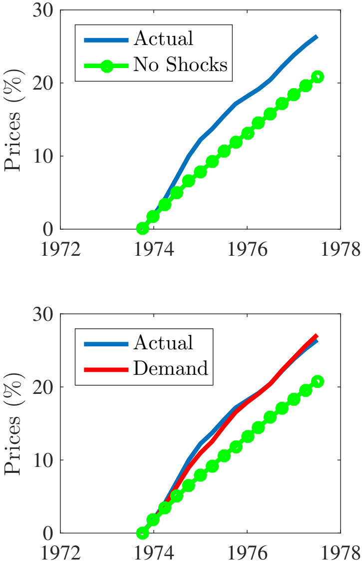
8. Results : Historical Decomposition

FIGURE 23 – First Oil Shock



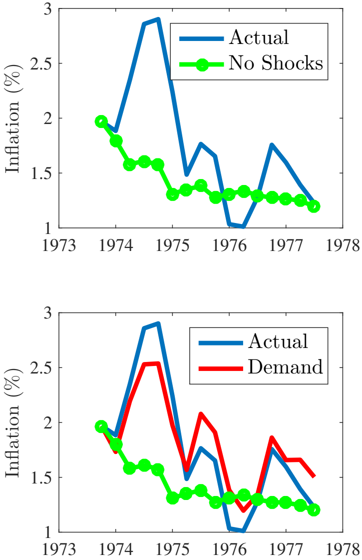
8. Results : Historical Decomposition

FIGURE 24 – First Oil Shock



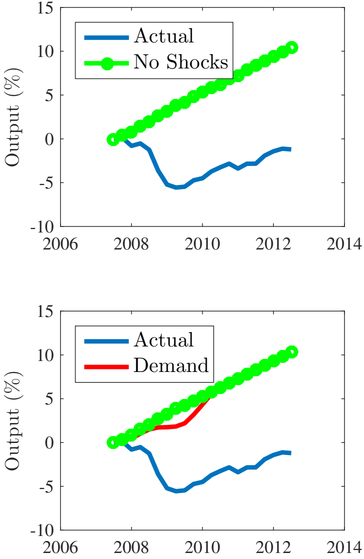
8. Results : Historical Decomposition

FIGURE 25 – First Oil Shock



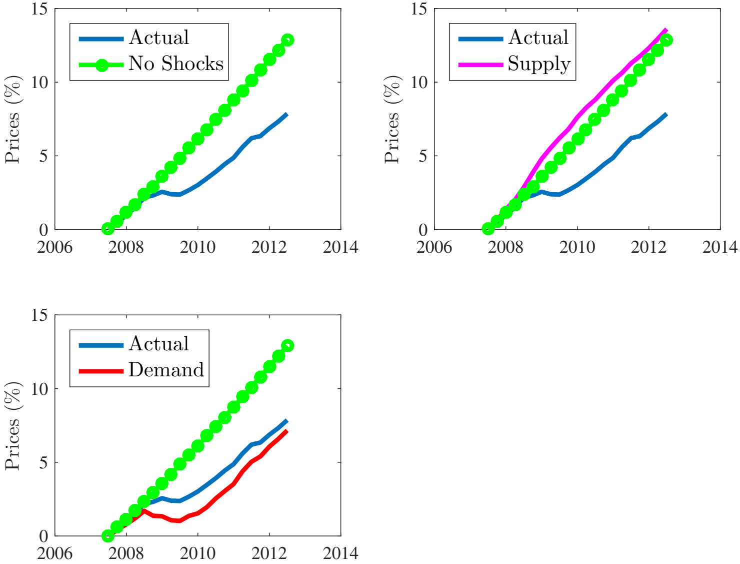
8. Results : Historical Decomposition

Figure 26 – The Great Recession



8. Results : Historical Decomposition

FIGURE 27 – The Great Recession



8. Results : Historical Decomposition

FIGURE 28 – The Great Recession